Are there Identifiable Trends in the Powerball?

A Micro and Macro Trend Analysis

W200 - Project 2
Emily Lieske and Sammantha Townsend

Introduction

During the Covid lockdowns of 2020, there was plenty of time to pursue new at home interests which for me involved playing the lottery. As I began to follow the numbers I began to recognize patterns that appeared more frequently than others. By doing these analyses manually it's easy to find statistical significance where none might actually exist as a result of small sample sizes.

This project aimed to explore trends in Powerball lottery numbers and develop a number generator grounded in statistical evidence. The initial question driving this effort was whether winning Powerball numbers displayed any discernible patterns or trends. This analysis aims to answer the question, "Are there actually identifiable trends in the Powerball?" To analyze this question, we were interested in evaluating to see if there were overall macro level trends in the Powerball in addition to micro level trends in the numbers drawn by investigating the following sub questions.

Sub Questions

Macro Level Trends:

- Has the amount of winnings gone up/gone down/stayed the same.
- Has the number or frequency of Jackpot winners changed?
- Are there any trends in the number of winners for other prizes?
- What is the average amount of time between jackpot winnings?
- Are there certain months where the jackpot amount is higher or times of the year there are more winners?
- Are there any seasonal trends (ie: higher amounts of winnings around the holidays etc)?

Micro Level Trends:

- Are certain numbers drawn more frequently than others?
- Are there any patterns in the ratio of odd/even numbers?
- Are there any patterns in the frequency of high/low number combinations?
- Are there any patterns in relation to the position of the numbers?

Data Collection:

To answer this, we began by gathering historical Powerball draw data from 2020 to 2025 using a custom-built web scraper. This data included details such as the date, jackpot amount, winning numbers, prize amounts and number of winners for each prize.

Data Set:

- https://docs.google.com/spreadsheets/d/1N-lyZY9rdT_DcnrzaW79gwOLEVKtfXdLjPY5RK5G9ME/edit?gid=0#gid=0

Data Cleaning:

The raw data contained inconsistencies and duplications that needed to be addressed. Duplicate rows were identified and removed, ensuring that each draw was represented only once in the dataset. Columns such as jackpot amounts were formatted inconsistently, often including symbols like "$" or text like "Million." These values were cleaned and converted into numeric form to allow for easier manipulation and analysis. Similarly, winning numbers, stored as strings, were parsed into structured lists, and date columns were converted to standard datetime formats.

For the micro level number analysis, we further cleaned the data to split out each number into its own column and converted these from strings to integers to better assess numeric trends. These steps ensured that the dataset was clean and ready for deeper analysis. In order to assess numbers into number groups or to determine if numbers were odd or even we applied functions to the data frames.

Powerball Analysis:

With a clean dataset, we conducted EDA to answer our central question: "Are there any trends in winning Powerball numbers?" The analysis focused on understanding historical patterns, jackpot distributions, and the behavior of winning numbers over time.

Macro Trend Analysis:

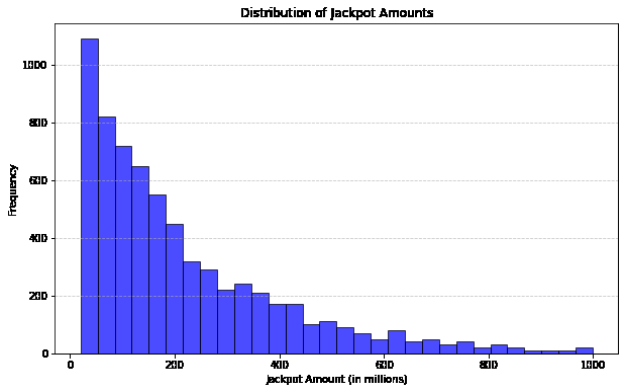
Over the past five years:

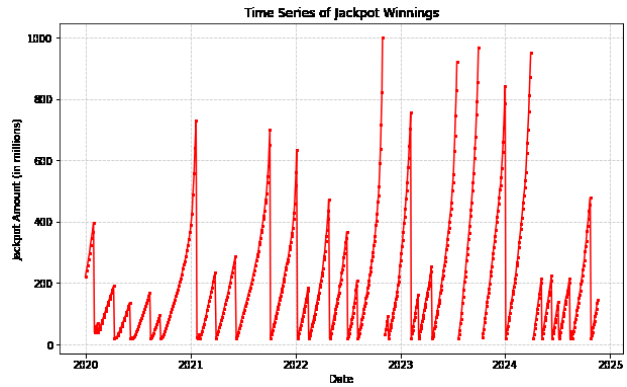
- There have been 687 Powerball drawings.
- There have been a total of 33 actual jackpot winners
- This gives each drawing about a 4.8% chance of there being a jackpot winner.
- On average there were about 6 Jackpot winners per year, this year being an outlier currently having 9 Jackpot winners.
-

Macro Level Jackpot Trends:

Analysis of Jackpot Amount Over Time:

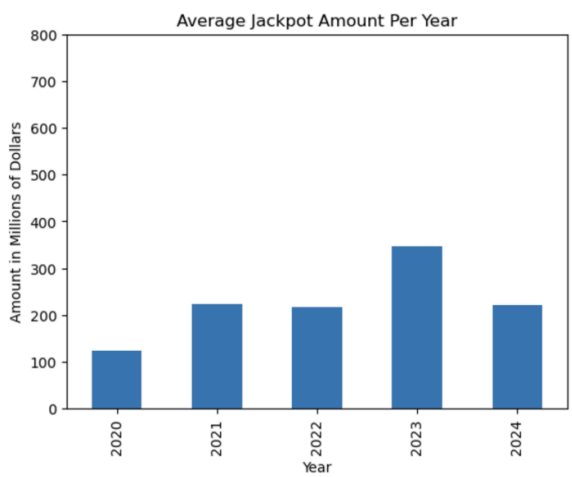
The histogram of jackpot amounts indicates that most jackpots are within the $10–200 million range, with a decreasing frequency as the amounts increase. Extremely high jackpots, such as those exceeding $500 million, are rare but do occur during extended rollover streaks.The time series plot of jackpot winnings highlights a pattern of sharp spikes corresponding to longer spans of time without a winner where the prize amount continued to increase.

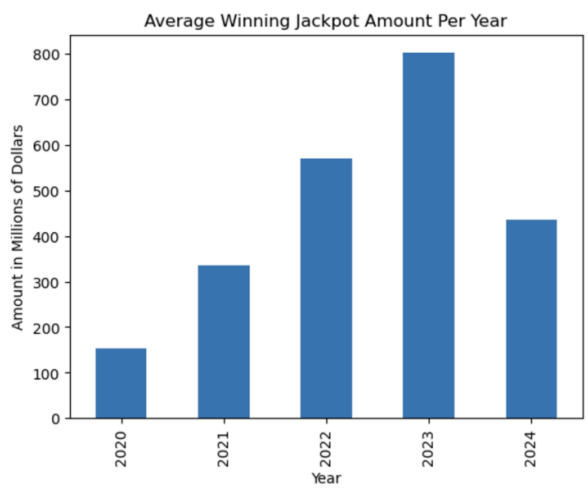




Looking at the time series it appears there was an increase in the size of Jackpots after 2021 and jackpots continued to rise throughout 2024 although the frequency of a winner remained consistent at about 6 Jackpot winners per year. This past year in 2024, we can see that the Jackpot amounts have drastically decreased but the frequency of winners has also increased by 50% with 2024 showing a total of 9 jackpot wins so far this year.

Analysis of amount of winnings



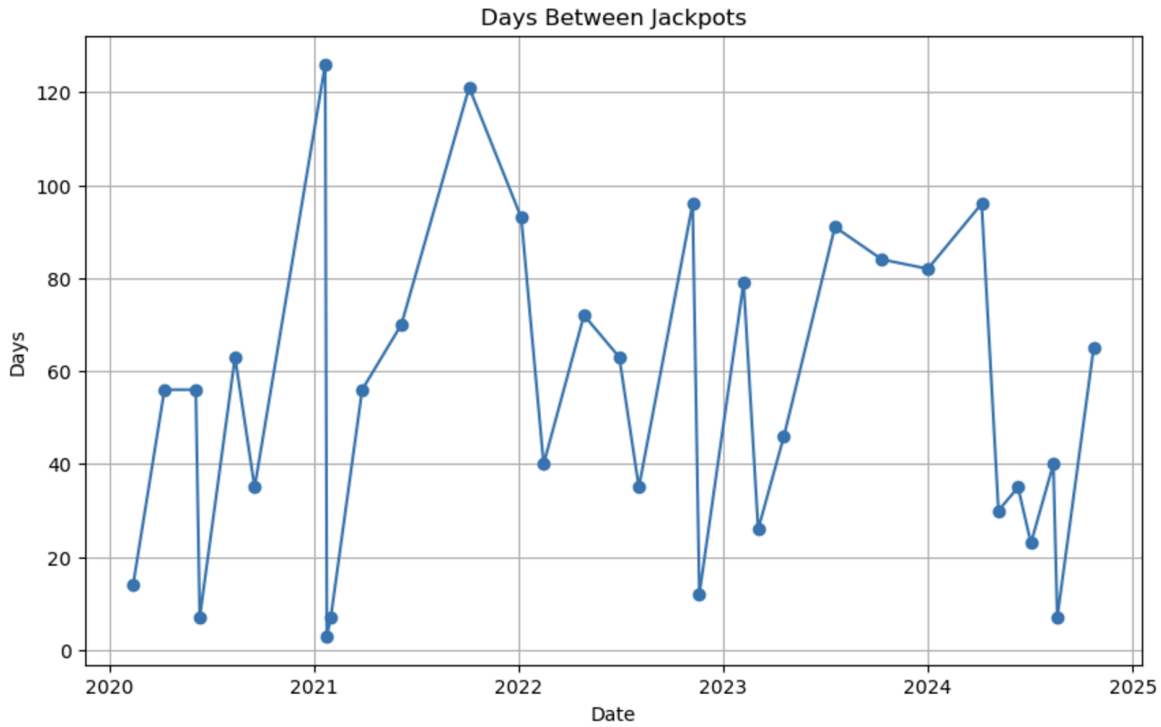


The bar chart of average jackpot amounts by year displays the average amount of the Jackpot per year for every drawing regardless of if there was an actual winner. Overall these amounts tend to average around the $250 million amount (with a couple of outliers above and below this amount). The first graph shows all the jackpot amounts (those with and without a winner) which skews the Jackpot amount downwards as it includes all the smaller amounts up to the final jackpot amount.

The second graph only evaluates the average Jackpot amount where there was an actual winner and the average amount of those jackpot winnings. Based on this analysis the average amount

that was actually won shows that the amount won has been steadily increasing since 2020. This year has been an outlier with more frequent small winners.

Analysis of time:



When we analyzed the average amount of time we were able to see that the lowest number of days for a jackpot winner was 3 days (meaning there was one back to back Jackpot winning since draws are conducted every three days). The longest length of time was 126 days for a Jackpot to be won. On average a Jackpot was won every 54 days, meaning that there are approximately six big Jackpot wins each year.

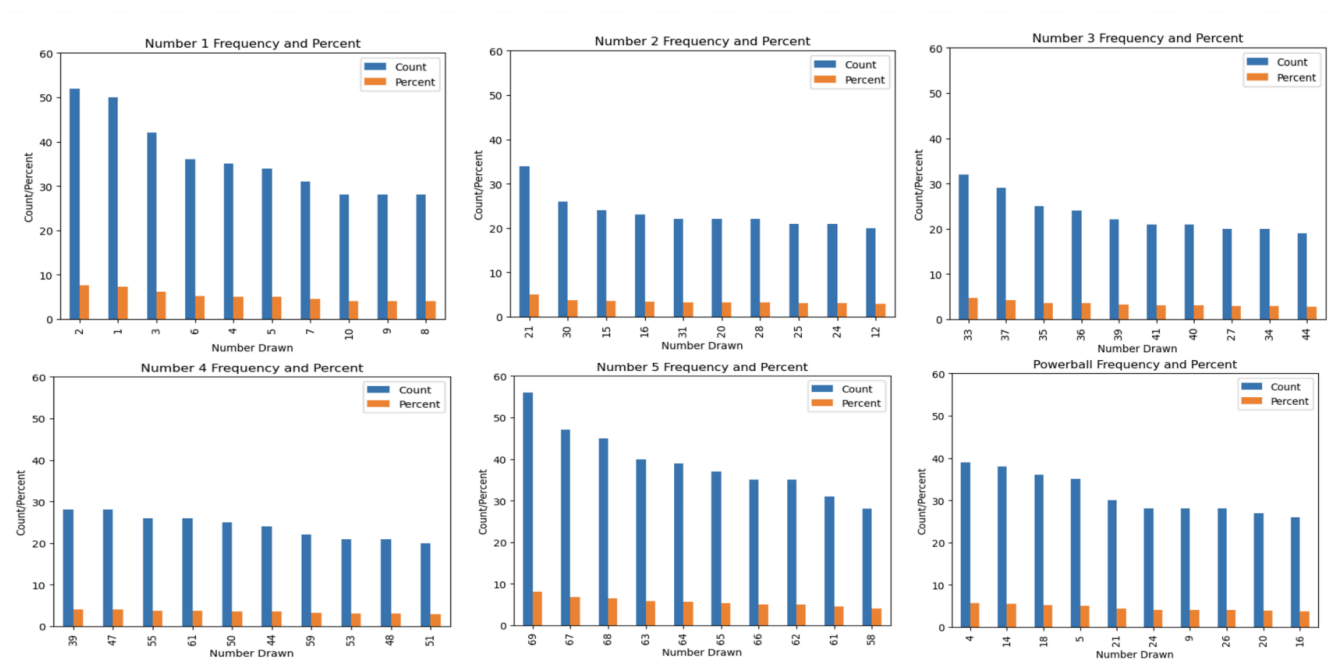
Seasonal Trend Analysis:

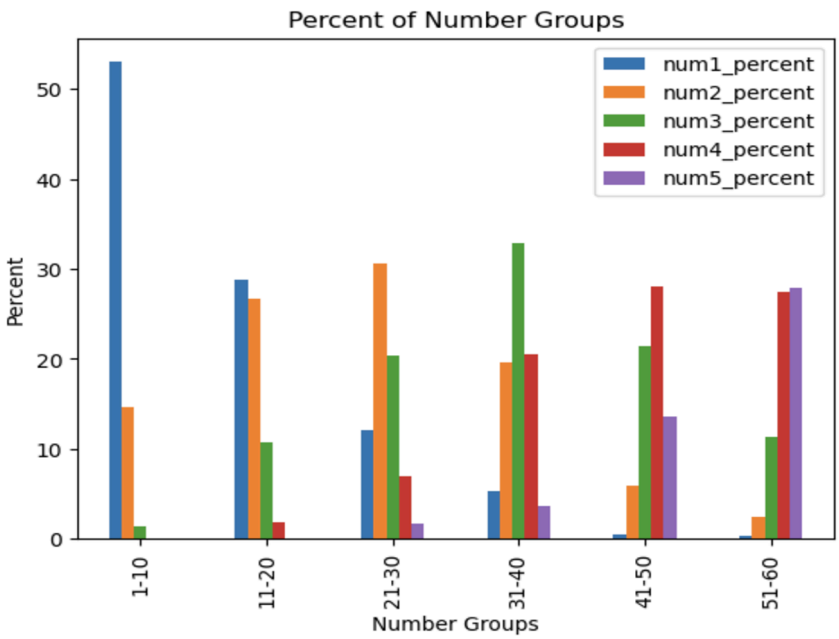
The average jackpot amount by month shows a clear seasonal trend, with higher jackpot amounts observed in the winter months, particularly around December and January. This suggests increased player participation during the holiday season, potentially driven by festive excitement and promotional activities. Conversely, lower jackpot amounts are seen in the summer months, indicating a possible decline in participation during this period. These seasonal fluctuations align with typical patterns of player behavior, where interest peaks during holidays and wanes during less eventful times of the year.

Micro Trend Analysis:
The powerball numbers are listed in order from lowest to highest (without knowledge of the order in which they were drawn). By analysing specific trends in the numbers location we were able to uncover some insights regarding the numbers in each position. As such we are able to identify some trends regarding the min, max and median numbers for each position.

Number Groups:
Simply by analysing the first number (the minimum) we were able to identify that numbers ranging from 1-10 made up the top ten spots for the first number, appearing in approximately 52.98% of all winning numbers. Additionally, numbers 60-69 appeared in 51.81% of all winning numbers. By selecting your numbers based on the likelihood of the numbers position we can attempt to use these weights to predict the likelihood of a number being represented from each group. The below table displays the frequency count along with the percent that each number group was represented in their respective position.

Number_Group	num1_count	num1_percent	num2_count	num2_percent	num3_count	num3_percent	num4_count	num4_percent	num5_count	num5_percent
1-10	364	52.98	101	14.70	10	1.46	NaN	NaN	NaN	NaN
11-20	198	28.82	183	26.64	74	10.77	13.0	1.89	NaN	NaN
21-30	83	12.08	210	30.57	140	20.38	48.0	6.99	12.0	1.75
31-40	37	5.39	135	19.65	226	32.90	141.0	20.52	25.0	3.64
41-50	3	0.44	41	5.97	147	21.40	193.0	28.09	93.0	13.54
51-60	2	0.29	17	2.47	78	11.35	189.0	27.51	192.0	27.95

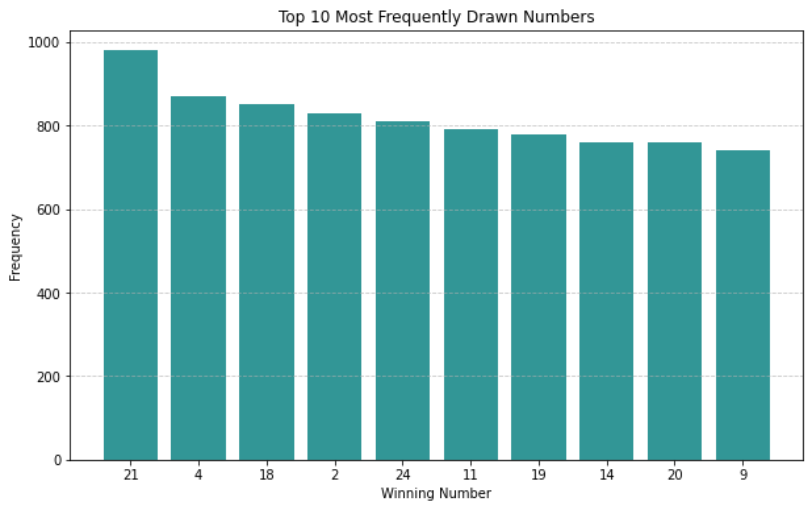




In this image we've split the numbers into 7 groups, each comprising a range of 10 numbers, for each position in the winning numbers you can see the associated percent numbers from that group were represented in that position.

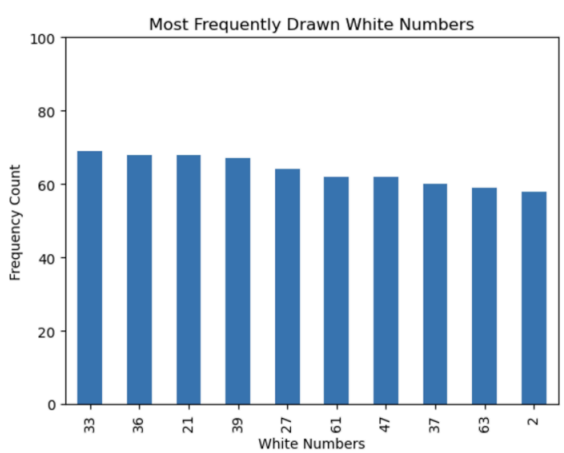
Using this analysis, if you were to pick numbers you'd likely want the first number to be between 1-10, the second somewhere between 1-20, the third between 11-30, the fourth between 31-40 and the fifth over 41 to increase the likelihood of having the winning number drawn. All the numbers in these ranges represent 50% of the total representation of that range of numbers in a given ball position.

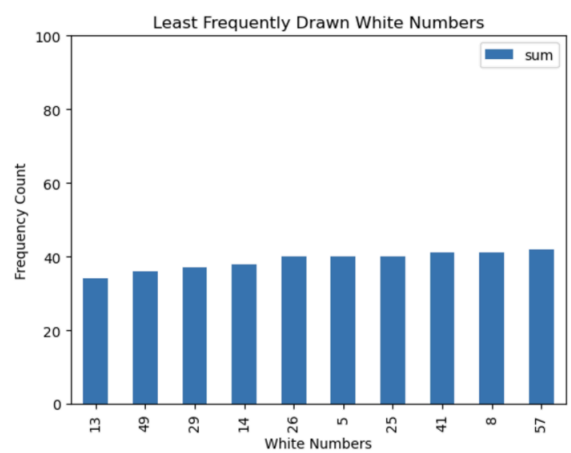
Frequently Drawn Numbers:
One of the most significant findings came from analyzing the frequency of individual numbers. A bar chart revealed that some numbers, such as 21, appeared far more frequently than others. The most frequently drawn numbers (which includes the white ball and the powerball), all appear to be under 26.



In order to conduct an analysis of the most frequently drawn numbers we had to assess the frequency of a number appearing as a white ball and the frequency of it appearing as the powerball separately. This is because the number 21 could be drawn once as the white ball and again as the powerball which resulted in the appearance that lower numbers are drawn more frequently (as seen in the above chart Top 10 Most Frequently Drawn Numbers). This statement is technically true since a number under 26 has the possibility of being drawn twice.

When assessing the white balls we can see that the following were the most frequently and least frequently drawn white balls:

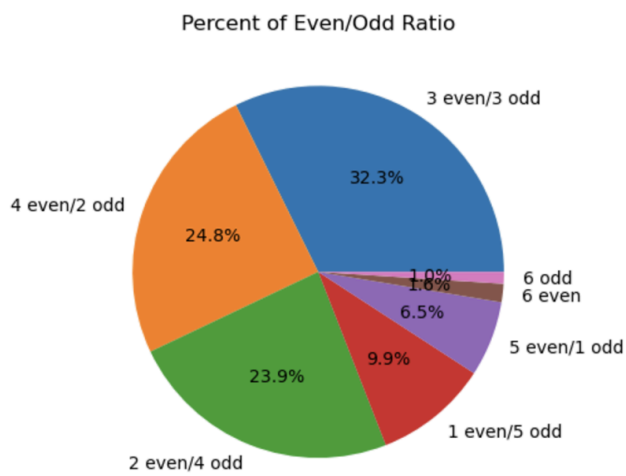


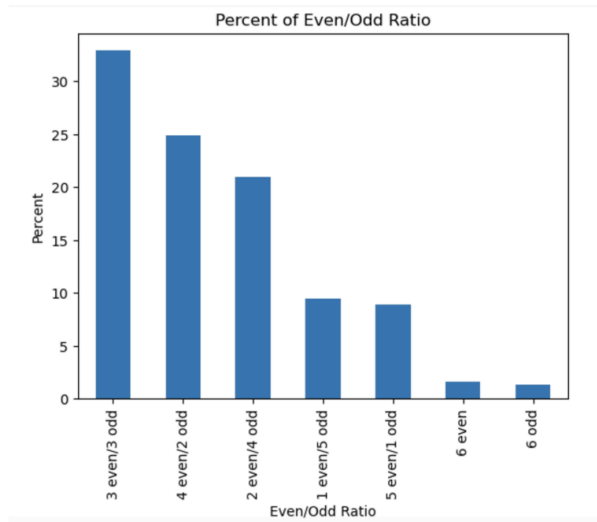


Based on this analysis, it appears that the number 33 is the most frequently drawn white ball being drawn 69 times out of the 687 winning draws we evaluated, meaning that the number 33 has a 10% likelihood of appearing in any drawing. Additionally, the number 33 was drawn nearly twice as often as the least frequently drawn number, 13, which was only drawn 34 times. As it turns out, there may be some anecdotal evidence to the superstition that 3 is a lucky number

and 13 an unlucky number. Additionally, when comparing the most and least frequently drawn numbers it supports the idea that some numbers are drawn more often than others. To see the most frequently drawn powerball numbers please refer to the Powerball Frequency and Percent table in the bottom right corner of the six graphs above.

Odd/Even Ratio:
We analyzed the data and discovered that numbers are most likely to have a 3 even/3 odd split (32.31%), followed by 4 even/2 odd (24.75%) and 2 even/4 odd (23.87%). When evaluating for these top three splits they accounted for 80.93% of all winning numbers (indicating that if you were to pick your own numbers you would probably want to avoid picking all even or odd numbers). There wasn't a significant difference between the likelihood of a number being odd or even based on its position, all five white balls had a similar probability of being odd or even (with a very slight 2% favorability towards even numbers).





Development of the Powerball Number Generator:

Building on the insights from our initial data analysis, the development of the Powerball number generator was guided by the dual goals of statistical accuracy and random variability. The generator was designed in Python and produces five white-ball numbers ranging from 1 to 69, along with one red Powerball number between 1 and 26. These ranges mirror the structure of actual Powerball draws. A key feature of the generator is its use of weighted probabilities for the white balls, ensuring that numbers with a higher historical frequency have a greater likelihood of being selected. This approach integrates empirical data into the generation process, making the numbers statistically informed rather than purely random.

The use of weighted probabilities was based on an analysis of historical draw frequencies. For example, numbers like 33, which appeared significantly more often in the dataset, were assigned

higher weights compared to less frequently drawn numbers like 13. This adjustment ensures that the generator reflects real-world trends while retaining the inherent unpredictability of lottery draws. Each weight was calculated to proportionally represent the frequency of appearance in the historical data, ensuring an accurate yet dynamic model.

Refining the Generator:

To further enhance the generator's outputs, several constraints were implemented based on patterns observed in our analysis. The generator ensures that at least one odd and one even number are included in each set of selected numbers, reflecting the common odd/even balance found in historical draws. This balance is critical for realism, as fully odd or fully even combinations were rare in actual data.

Another refinement was limiting consecutive numbers in the output, as long sequences of adjacent numbers were also uncommon in historical draws. These constraints ensure that the generator produces plausible outputs that align with observed patterns without overfitting to specific sequences or trends.

However, initial versions of the generator included overly restrictive rules, such as requiring historically common pairs of numbers. While this improved statistical alignment, it often caused the generator to fail in producing valid combinations within allowable attempts. To address this, these constraints were relaxed in subsequent versions, striking a balance between adherence to trends and the flexibility needed for randomness.

The red Powerball number was treated differently, as our analysis did not uncover significant trends in its frequency distribution. Therefore, the red Powerball is generated purely randomly within its range of 1 to 26, ensuring variability while maintaining statistical neutrality.

Iterative Development and Debugging:

The development process involved iterative refinement to overcome technical challenges and improve the generator's reliability. Early versions of the generator used a recursive algorithm to apply constraints, but this approach frequently led to stack overflow errors when constraints couldn't be satisfied. This was particularly problematic in cases where highly restrictive rules severely limited the pool of valid outputs.

To resolve these issues, the recursive algorithm was replaced with an iterative approach. This new method employed a retry limit to avoid infinite loops and ensured that the generator could produce valid combinations efficiently. By systematically debugging and refining the code, we optimized the generator to balance statistical grounding with operational flexibility.

Another key improvement was the introduction of modularity in the code, allowing for easier adjustments to weights and constraints. This adaptability ensures that the generator can be updated with new data or altered to reflect changing trends in Powerball draws.

Final Output and Application:

The final version of the generator successfully combines historical insights with randomness, producing statistically informed yet varied outputs. For instance, it might generate white-ball numbers such as [4, 15, 23, 36, 42] and a red Powerball of 12. Each run of the generator provides a unique combination, ensuring sufficient variability while adhering to the patterns observed in historical draws.

This generator not only answers the question of whether trends exist in Powerball draws but also demonstrates how data science can be applied to practical, real-world scenarios. Although the lottery is inherently unpredictable, the generator serves as an educational tool, highlighting the interplay between randomness and statistical patterns. By exploring these trends, we can gain insights into the mechanics of lottery draws and the power of data-driven decision-making, even in highly random systems.